

Special Seminar on Multi-Modal AI

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<https://cis7000-fall26.thelabone.org/>

Lecture 8: Speech and Audio Representation

Special Seminar on Multi-Modal AI

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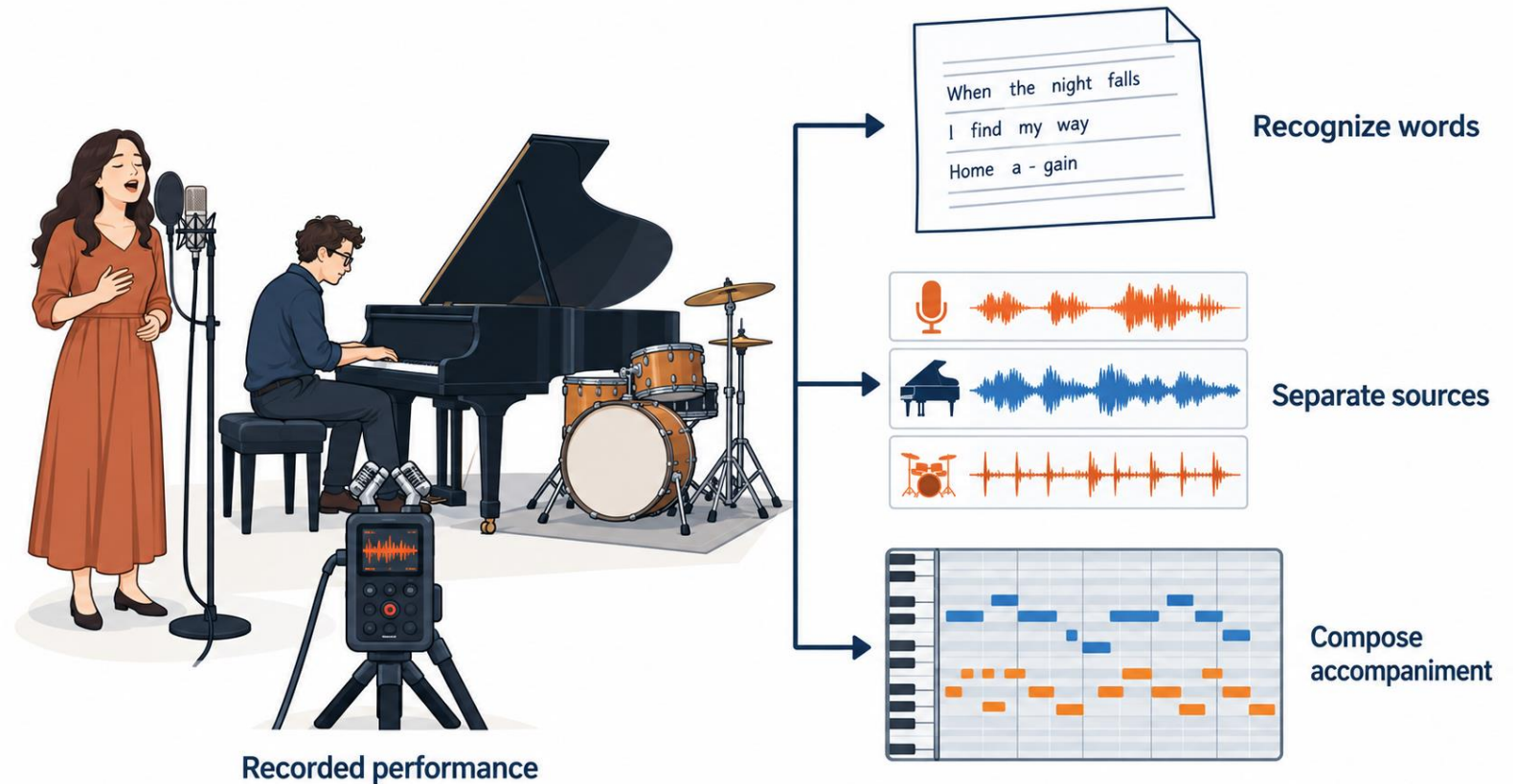
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Basic Tasks in Audio Understanding

- Recognize what was said, recover each source, or compose a new accompaniment



Today's Agenda

- Principles of Audio Representations
- Principles of Audio Processing Units
- Principles of Audio Representation Learning Algorithms
- Principles of Audio Task Evaluation
- Symbolic Representations for Music

Today's Agenda

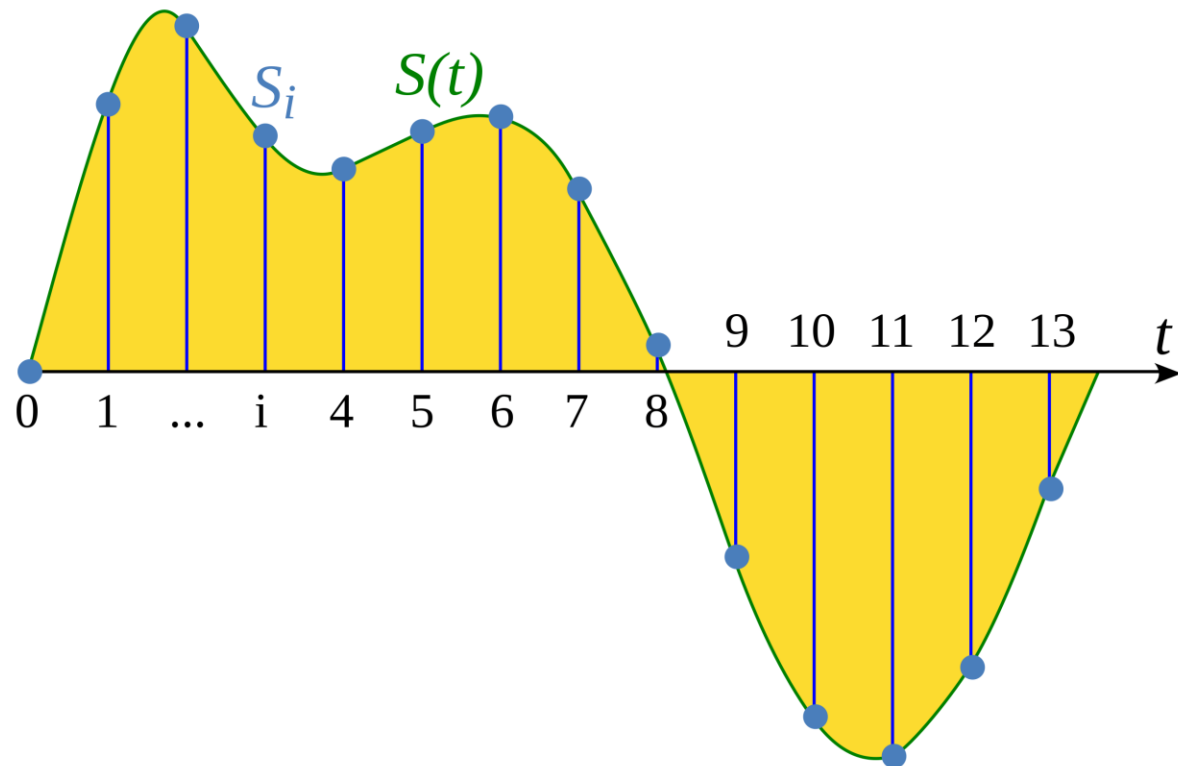
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Audio Representation Basics

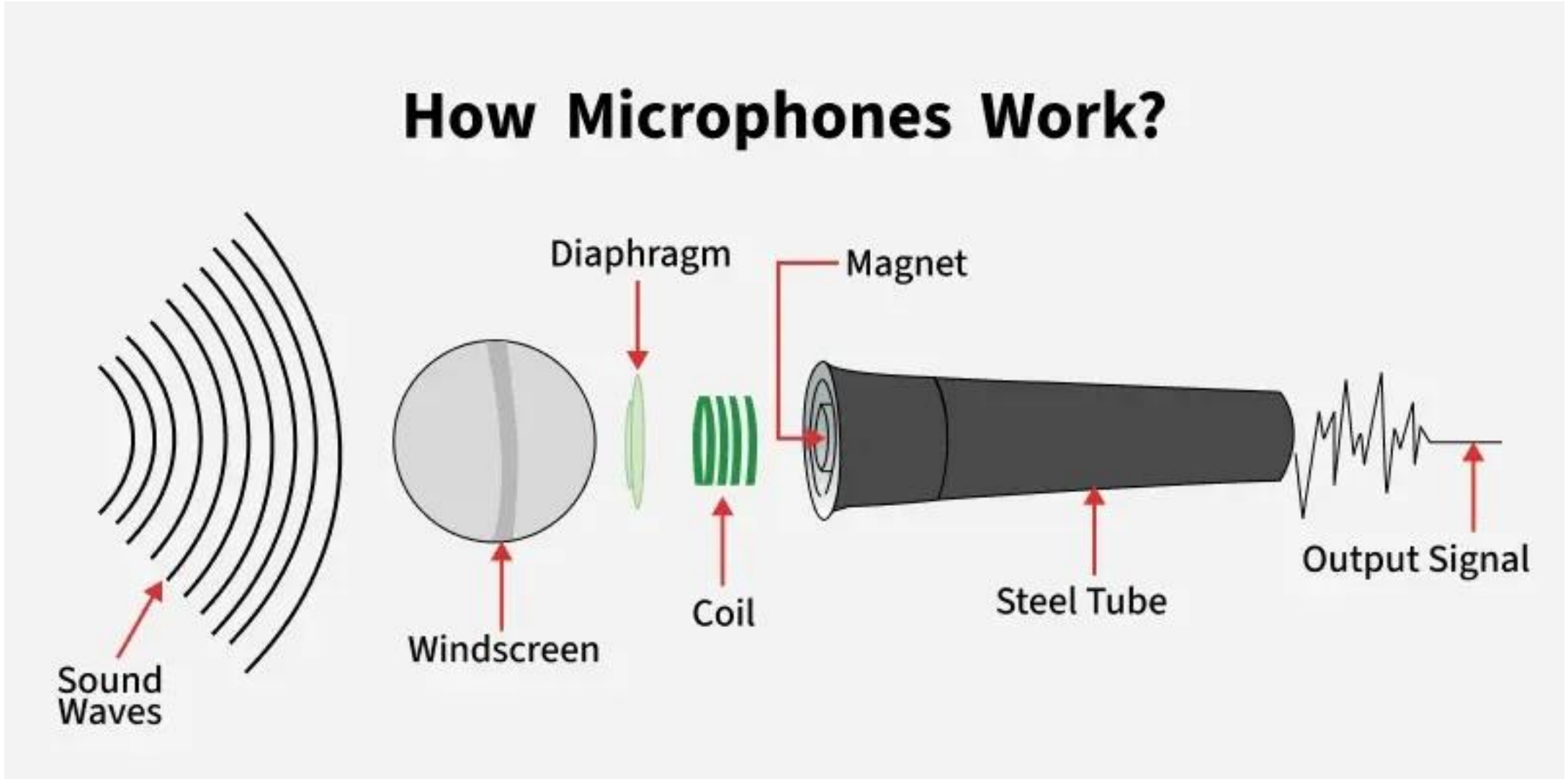
- First: waveform, spectrum and spectrogram
- Next: mel filtering, logarithms and MFCCs

Waveform = Measurement of Pressure

- A waveform is the recorded amplitude at each timestamp
- Sampling rate = time spacing; sample values can be positive or negative
- At 16 kHz, one second contains 16,000 samples

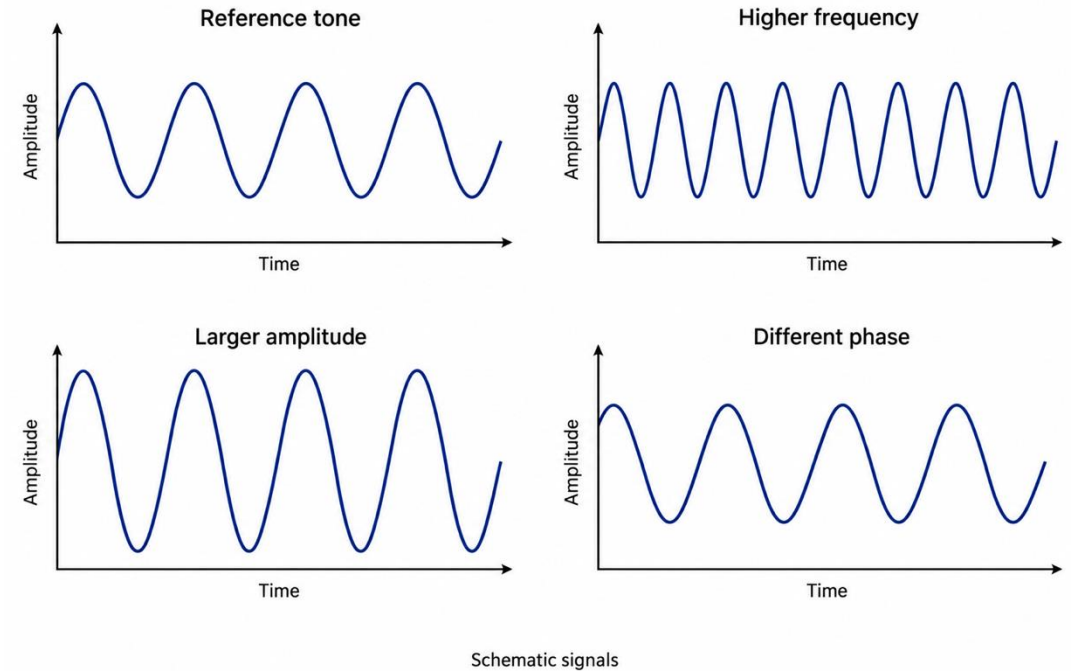


Waveform = Measurement of Pressure



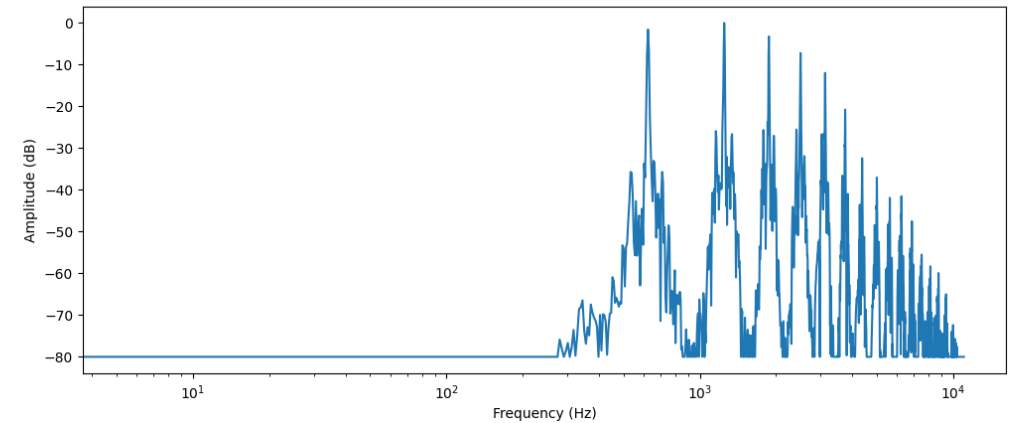
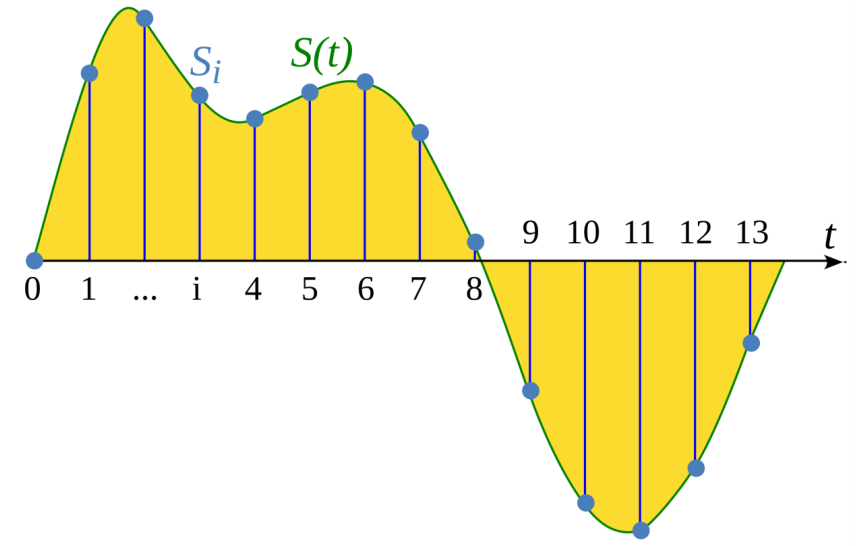
The Frequency Perspective

- Frequency: how many cycles occur per second
- Amplitude: the size of the oscillation
- Perceived pitch = Frequency
- Perceived Loudness = Amplitude



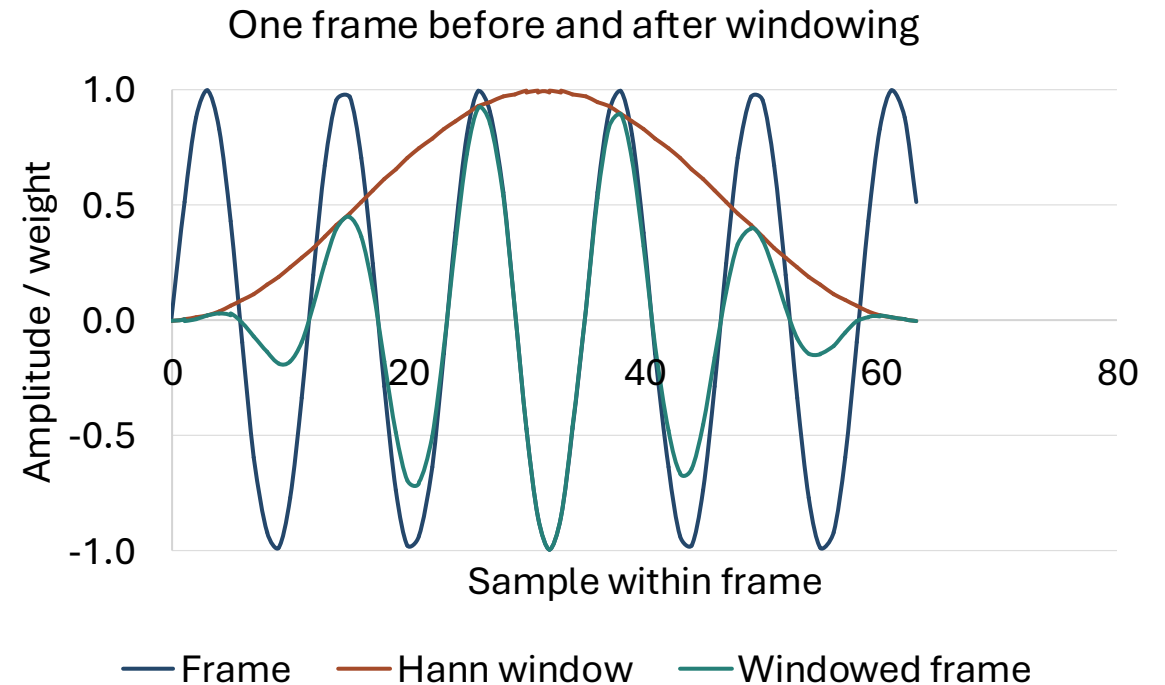
Spectrum Representation

- A Fourier transform decomposes one **segment** into oscillations
- Horizontal axis: frequency; vertical axis: magnitude of each component



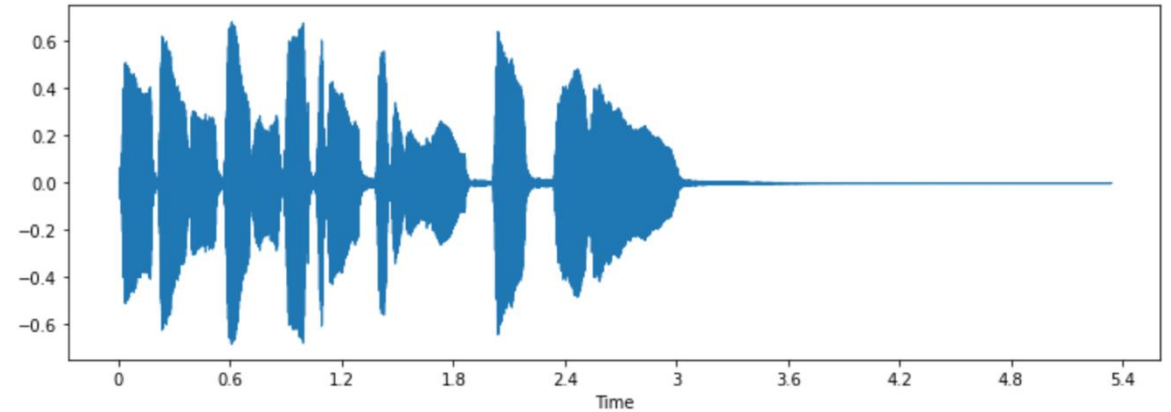
Framing and Windowing

- Take short frames and advance by a fixed hop
- Window weights adjust the weights per step in the frame
- Compute a Fourier transform of the weighted samples



STFT: Windowing at Each Frame

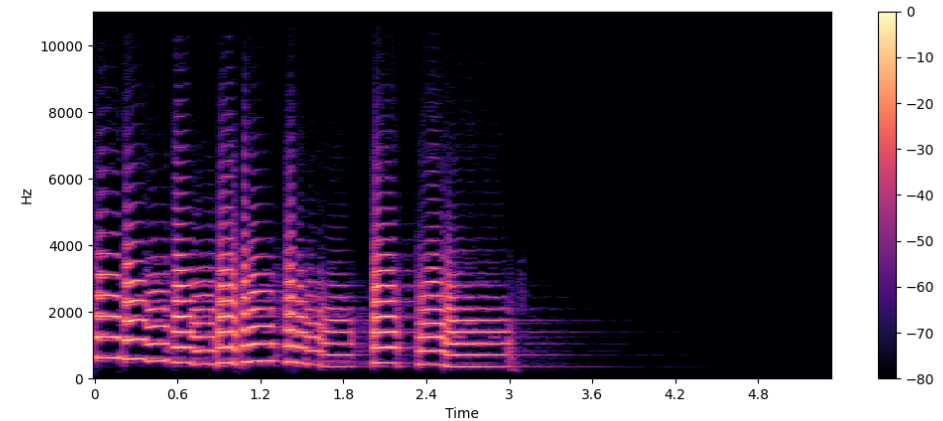
- Window each overlapping frame, then compute its Fourier transform
- Each transform returns complex coefficients indexed by frequency



$$X[m, k] = \sum_{n=0}^{N-1} x[n + mH]w[n]e^{-j2\pi kn/N}$$

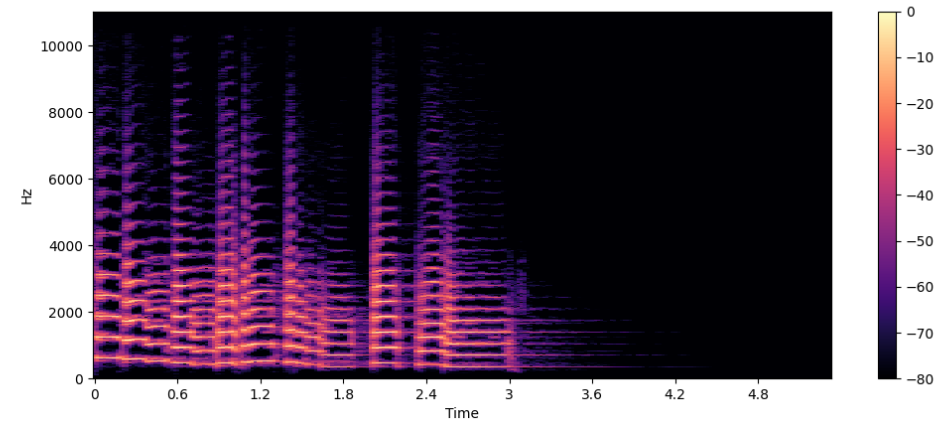
Spectrogram

- Horizontal axis: time; vertical axis: frequency; color: magnitude or power
- A vertical slice shows one frame's spectrum, often displayed in dB



The Harmonics Structure

- A periodic source often produces energy near integer multiples of a fundamental
- The harmonic envelope helps distinguish voices and instruments at the same pitch
- Drums, fricatives, and inharmonic sounds have other structure
- Longer frames separate nearby frequencies but blur timing



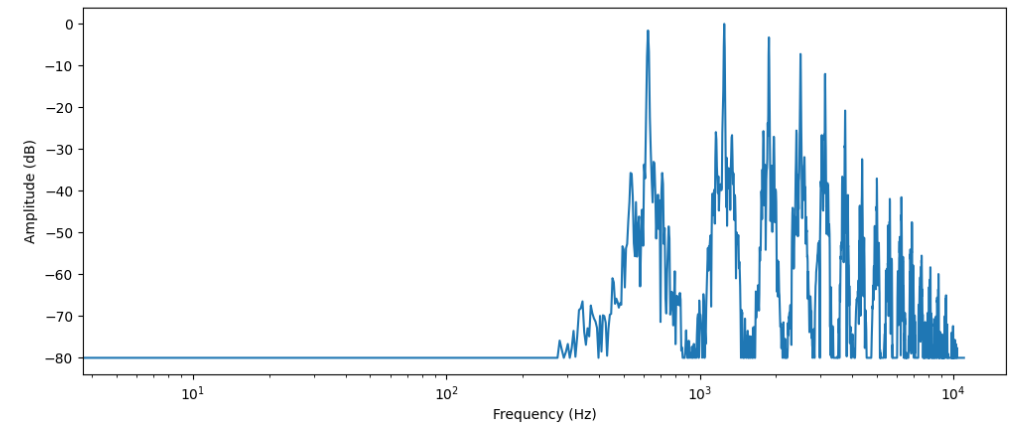
Magnitude and Amplitude

- Magnitude takes the absolute value of a complex coefficient

$$P = |X|^2$$

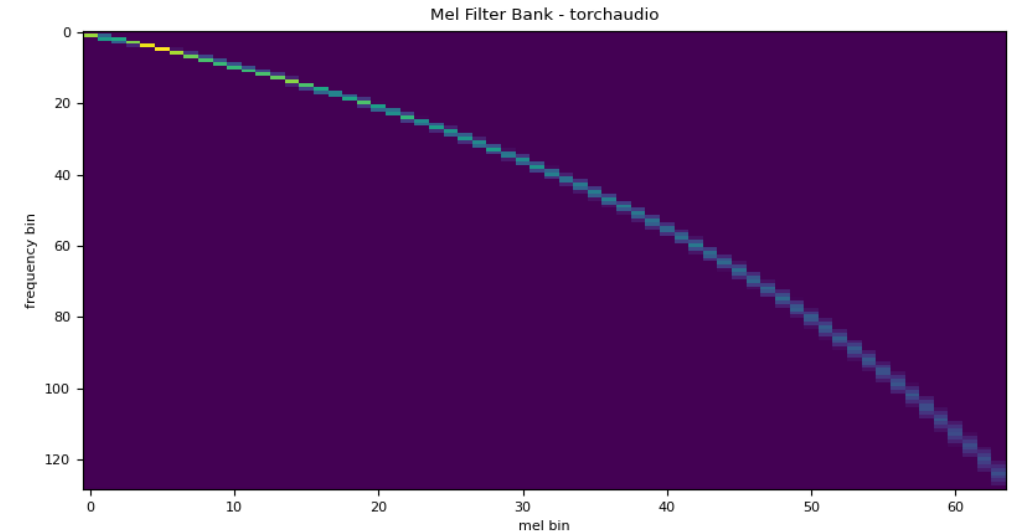
$$dB = 20 \log_{10} \frac{|X|}{A_{ref}}$$

- A factor of two in amplitude is about +6 dB



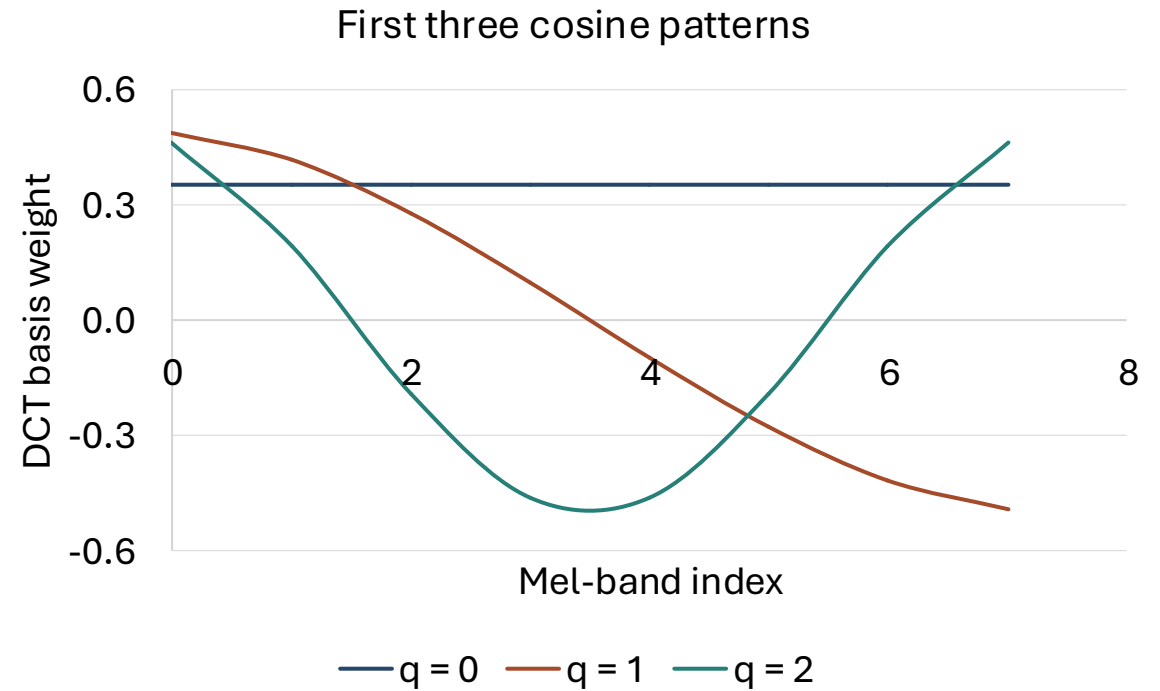
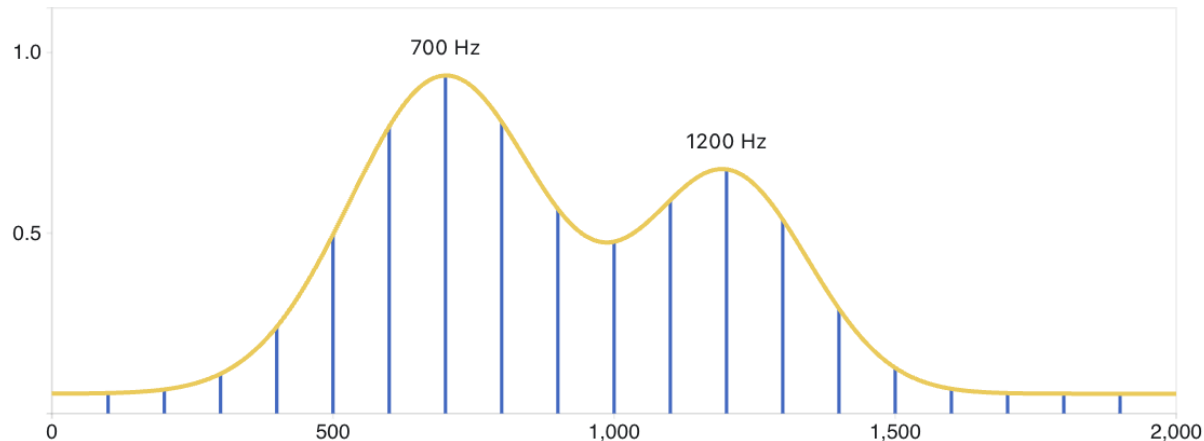
Mel Filterbanks: Pooling

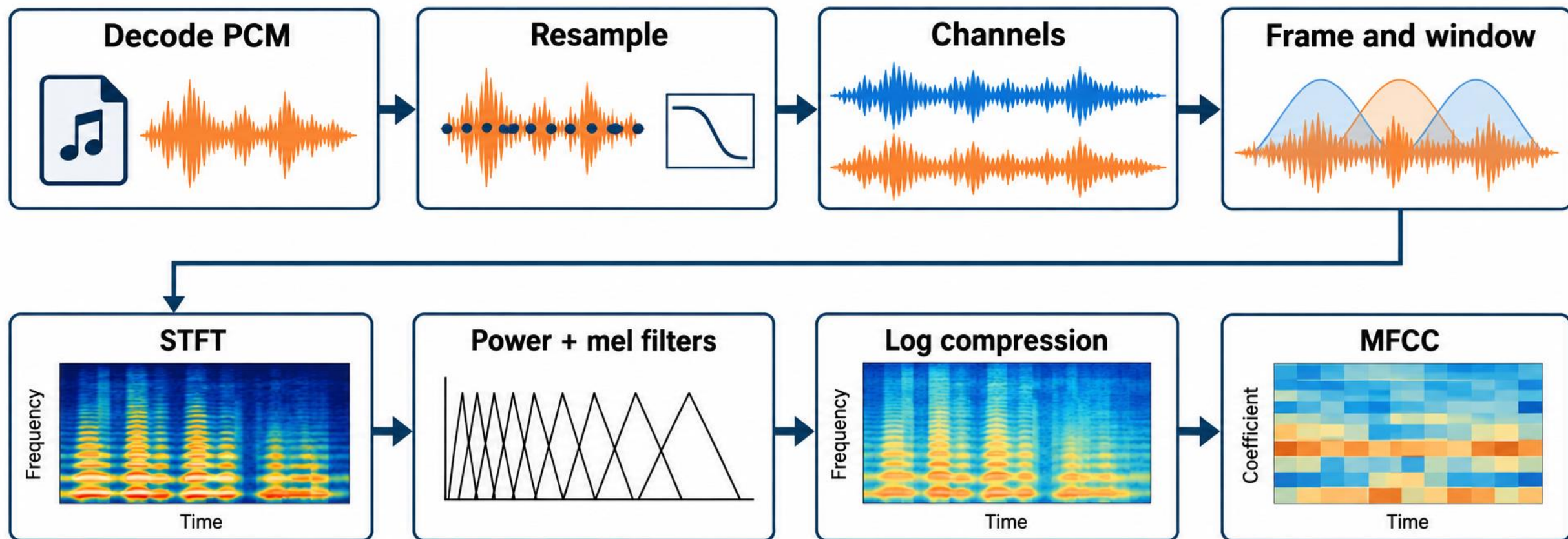
- Triangular filters weight and sum nearby spectral powers
- Broader bands at high frequencies
- A mel spectrogram uses mel-band index in place of linear frequency

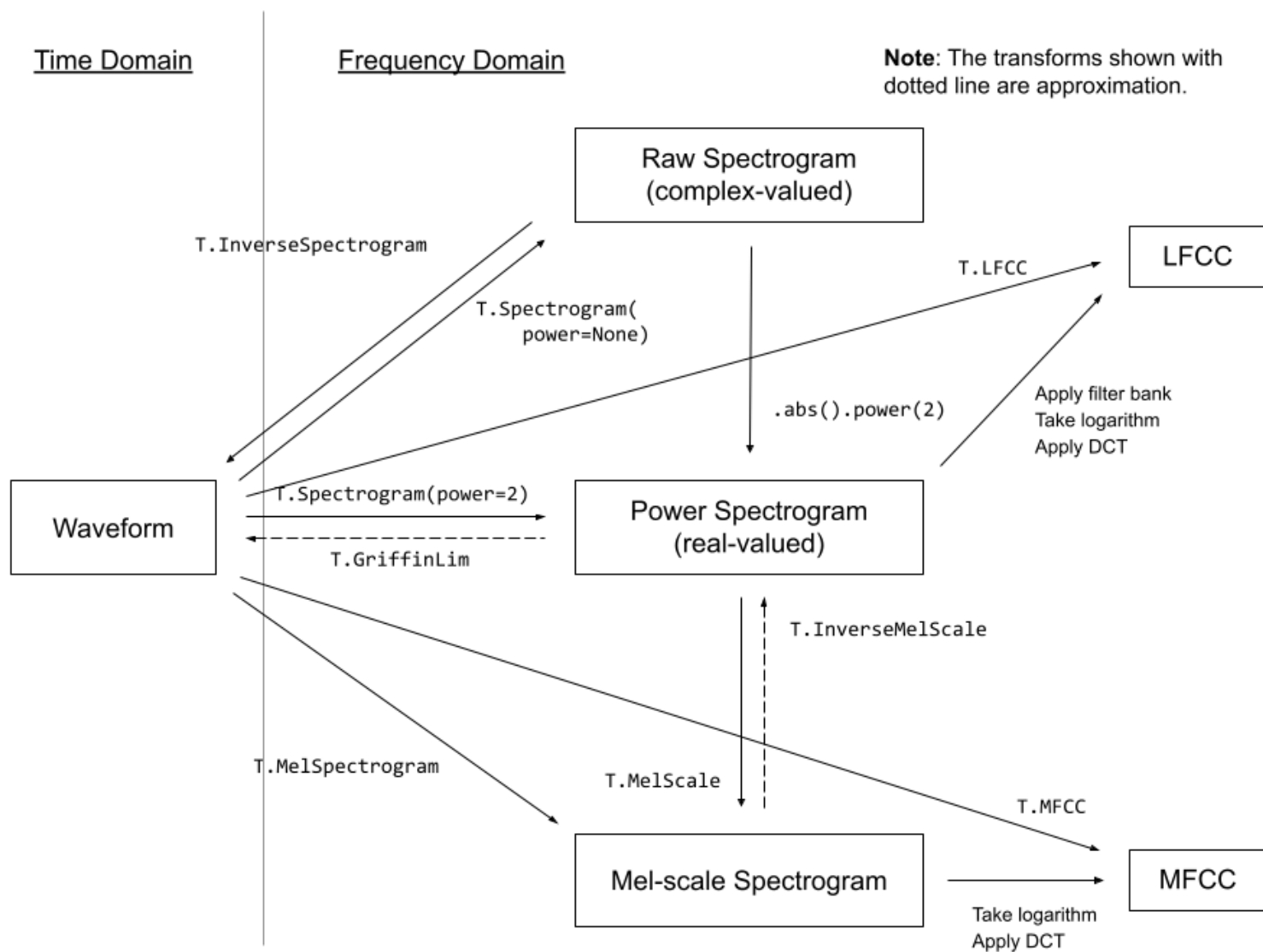


DCT for Compression

- Match cosine patterns across log-mel bands
- Low orders describe broad spectral shape
- Transform the band axis within each frame







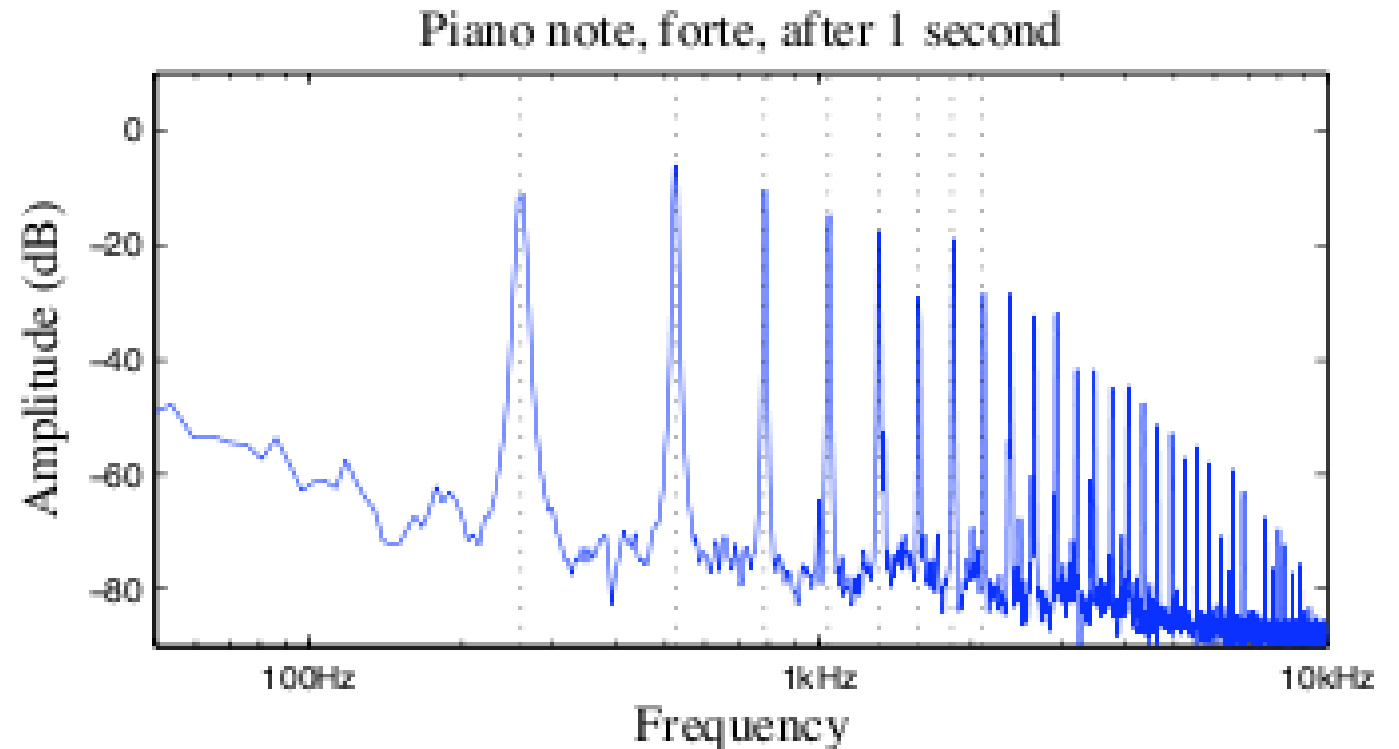
MFCC: one complete numerical recipe

- For one second of 16 kHz mono audio, keep 13 coefficients per frame
- Changing representation changes the meaning of the feature axis

Operation	Output shape	Meaning of one column
Waveform	$16,000 \times 1$	Sample amplitude
Frame + window	98×400	Sample within a frame
Real FFT + power	98×257	Linear-frequency bin
Mel pooling + log	98×40	Log mel-band energy
DCT	98×13	Cepstral coefficient

Why LOG everywhere?

- Our sense of pitch is log-scale
- Our sense of loudness is log-scale
- Harmonics is log-scale



Why LOG everywhere?

Weber–Fechner law

Article [Talk](#)

From Wikipedia, the free encyclopedia

The **Weber–Fechner laws** are two related [scientific laws](#) in the field of [psychophysics](#), known as **Weber's law** and **Fechner's law**. Both relate to human perception, more specifically the relation between the actual change in a physical [stimulus](#) and the perceived change. This includes stimuli to all senses: vision, hearing, taste, touch, and smell.

[Ernst Heinrich Weber](#) states that "the minimum increase of stimulus which will produce a perceptible increase of sensation is [proportional](#) to the pre-existent stimulus," while [Gustav Fechner](#)'s law is an inference from Weber's law (with additional assumptions) which states that the intensity of our sensation increases as the [logarithm](#) of an increase in energy rather than as rapidly as the increase.^{[\[1\]](#)}

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Convolutions on Spectuograms

- Motivation: denoising
- Input: Spectrograms
- Research question: which architectural bias suits audio

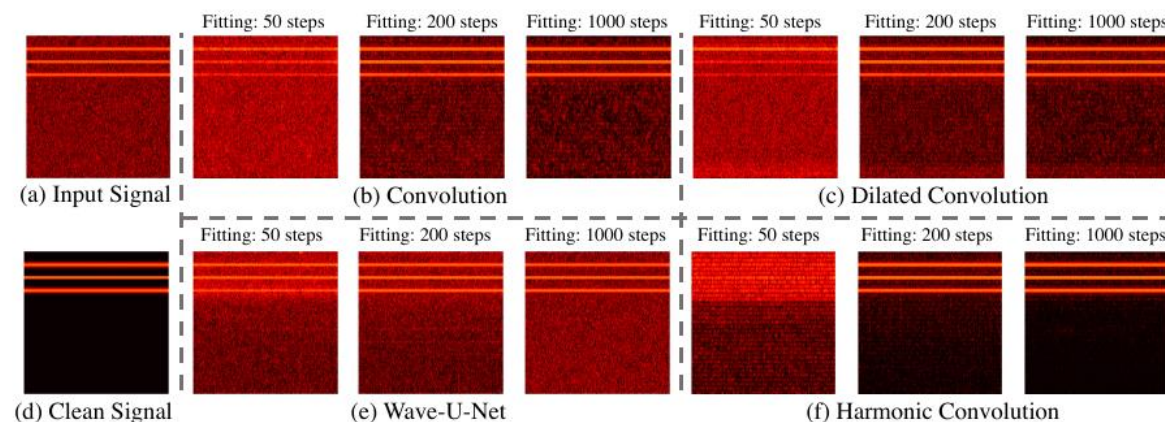
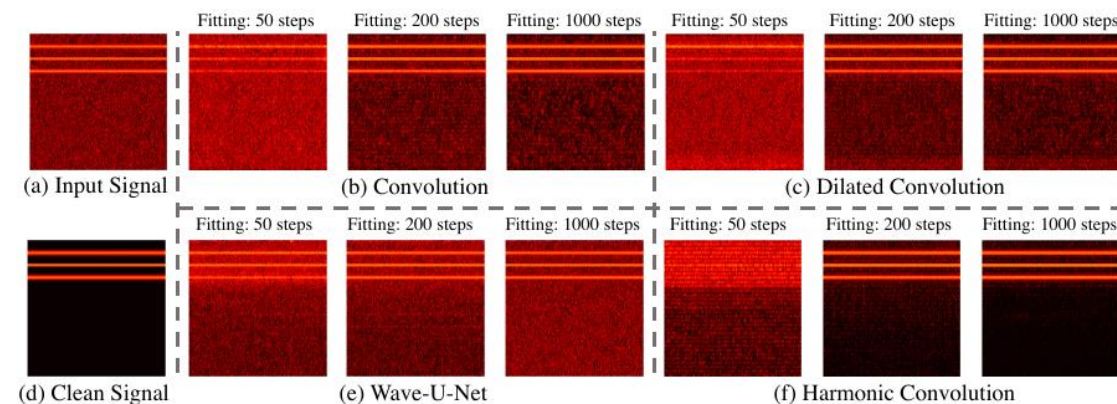


Figure 1: A simple illustrative case: fitting a harmonic sound corrupted by Gaussian noises. The detailed setup of all networks can be found in the appendix. (a) The input to networks is a harmonic sound composed by sinusoids of 1,000, 2,000, and 3,000Hz, plus random Gaussian noises. (d) The clean, target signals. They are shown here as references, not accessible to any models. (b)(c)(e) Both temporal and spectral-temporal convolution networks start with a very noisy output at 50 iterations; they then fit the signal as well as the background noises at 200 iterations. At 1,000 iterations, the output is similar to the one at 200 iterations. This suggest that these networks fit noises and the target signal simultaneously. (f) Harmonic Convolution starts with a noisy output as well, but at 200

The “Prior” from An Architecture

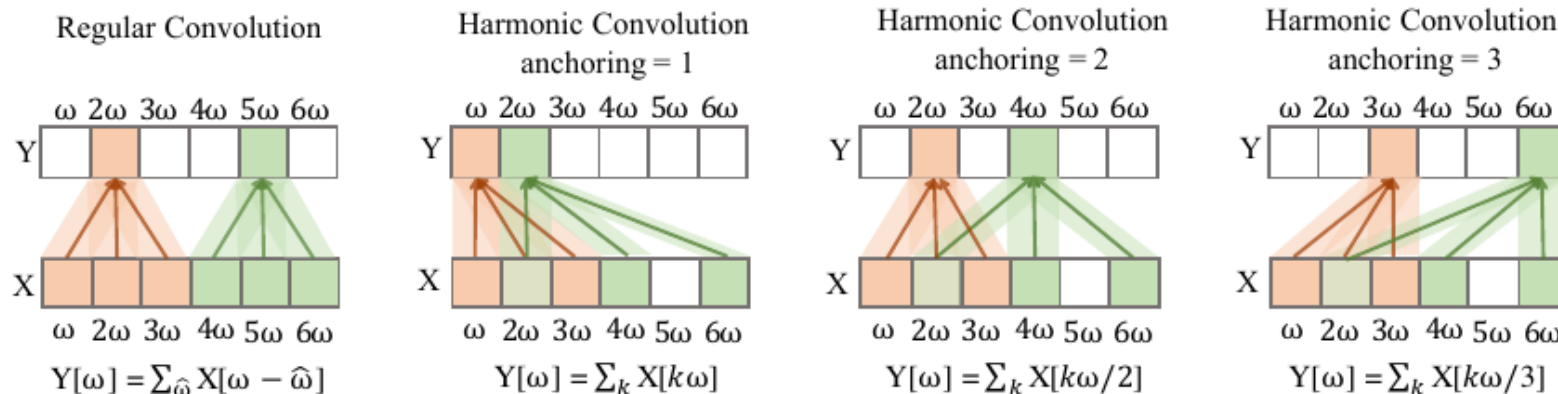
- Input: clean signal + noises
- Learn network weights to reproduce the observed input
- This visualizes the prior of “what are possible audio data”



$$\min_{\theta} \| f_{\theta}(z) - X_{observed} \|^2$$

Harmonic Convolution

- Issues with Convolution: adjacent frequency bins do not belong to the same "source"
- Intervention: aggregate at harmonic multiples



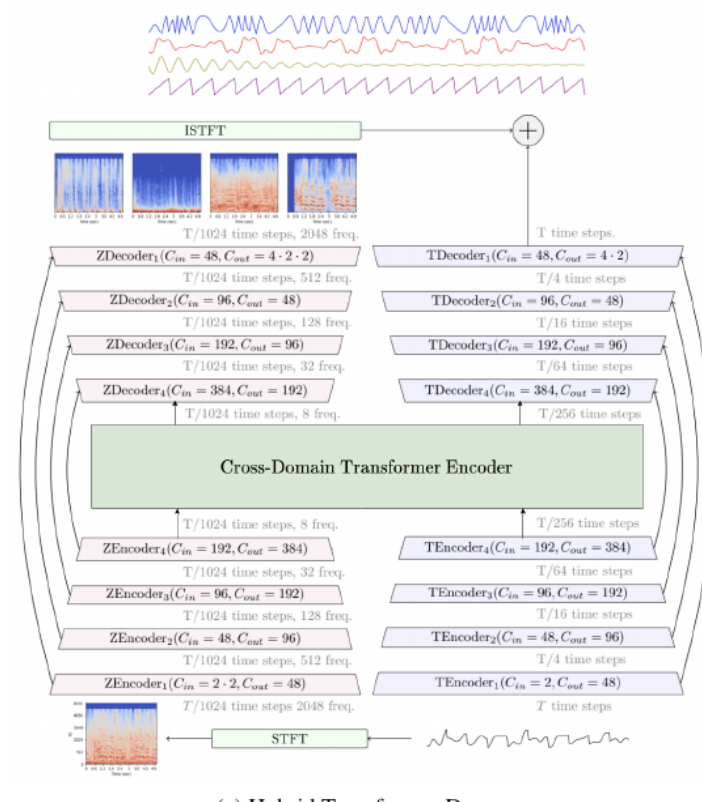
Restoration evidence and its limits

- Harmonic kernels improve denoising
- Limitation: harmonic structure is less informative for noisy or inharmonic sources

Kernel	Speech SSNR	Music SSNR
Regular	4.13 dB	5.07 dB
Dilated	4.85 dB	5.38 dB
Harmonic	7.12 dB	9.85 dB

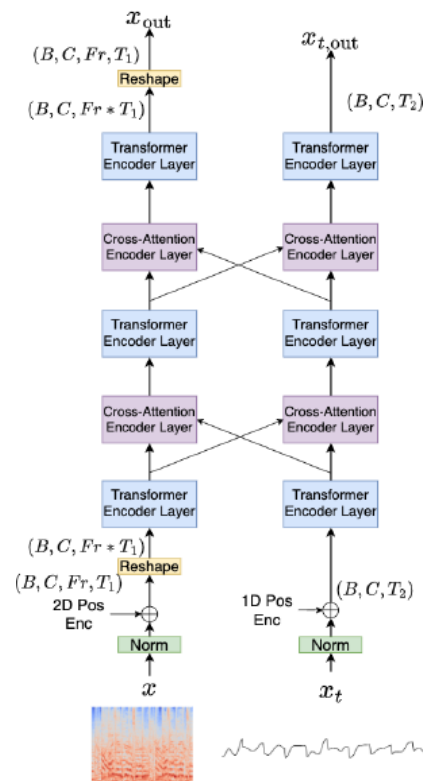
Waveform vs Spectrogram Models

- Spectrogram masking exposes harmonic patterns, but may blur multiple sources
- Wave-U-Net and Demucs learn directly in the waveform domain
- Hybrid Demucs combines both views



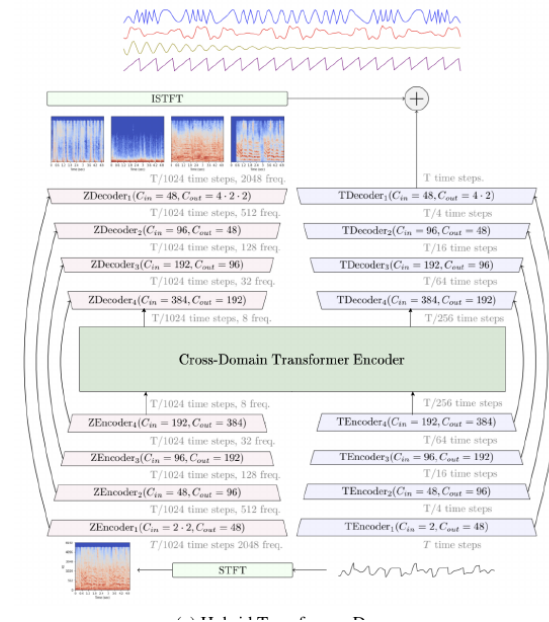
HT Demucs: the key idea

- Idea: local evidence may not identify an overlapping instrument
- Separate spectral and waveform encoders retain complementary structure
- Cross-domain attention shares context before decoders predict four stems



Training and Inference

- Input: stereo mixtures at 44.1 kHz; targets: aligned source waveforms
- Train with waveform L1 loss, stem remixing, and pitch/tempo augmentation



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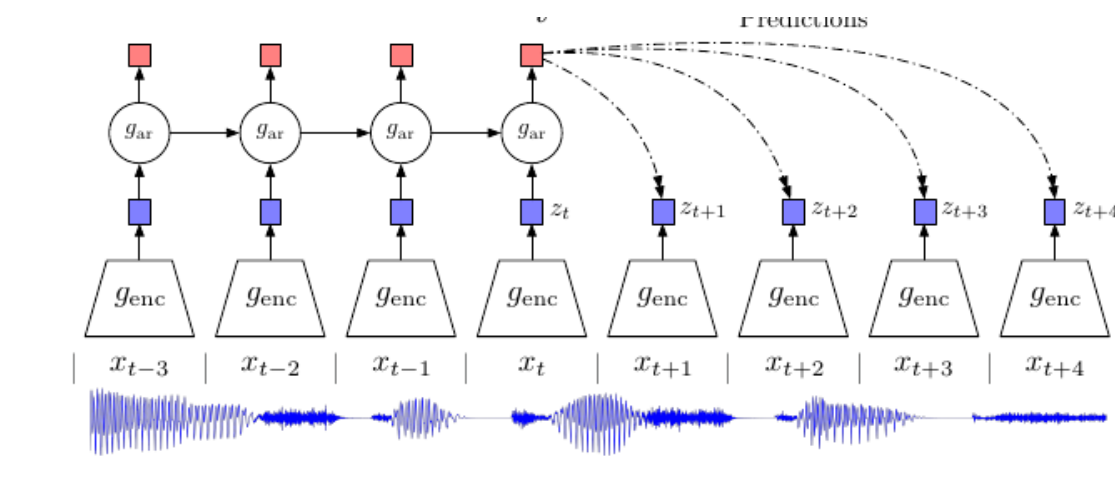
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Representation Learning in Audio Data

- Based on the MFCC Feature
- Next-token prediction: CPC
- Masked language model: wav2vec 2.0
- Key problem: Audio signals do not have a predefined “token”
- But the underlying structures are close: music notes, speech parts

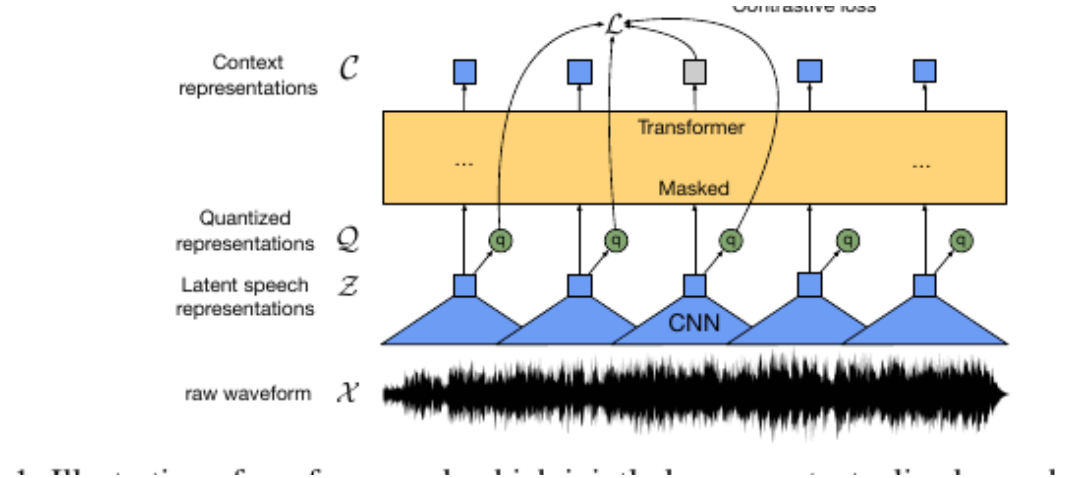
CPC: Predicting Future Latent Features

- Encode speech and summarize the past
- Identify real future latents among sampled negatives



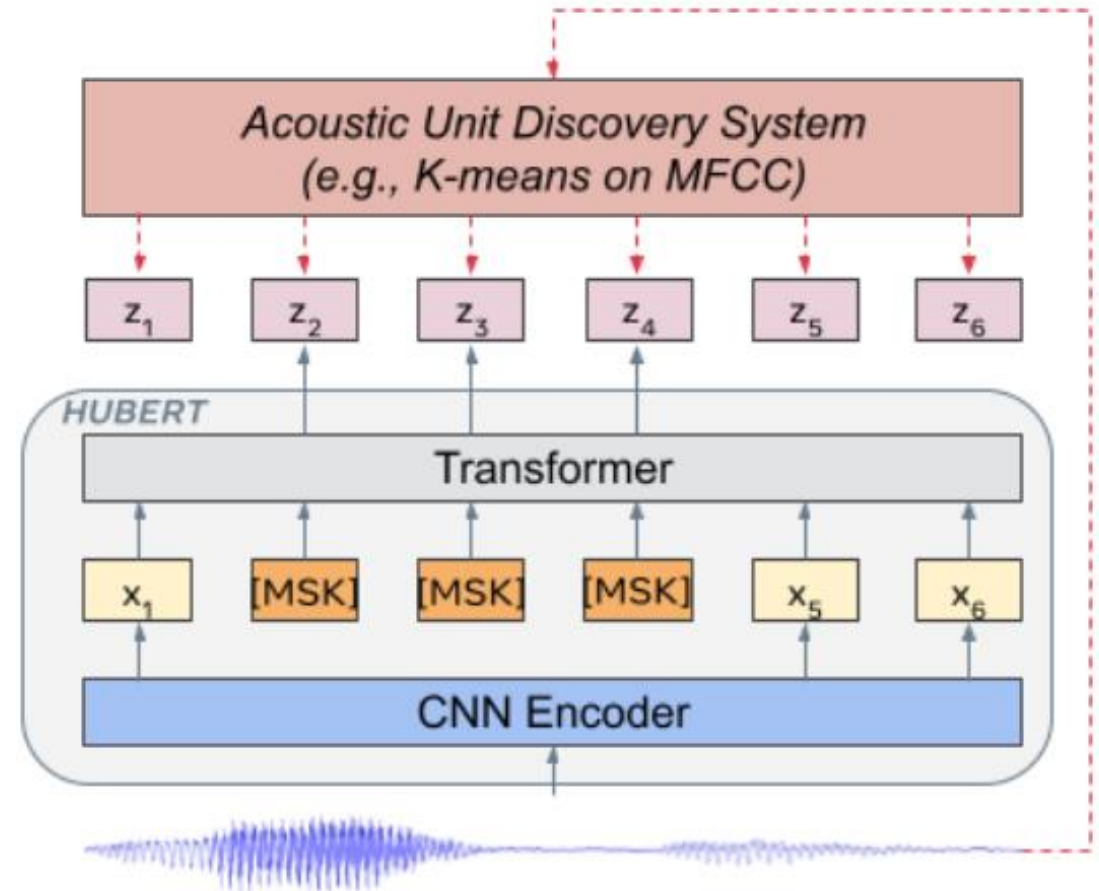
wav2vec 2.0: masked contrastive learning

- Use masked context to identify quantized targets
- Learn targets jointly with a codebook diversity loss
- HuBERT instead predicts offline cluster labels



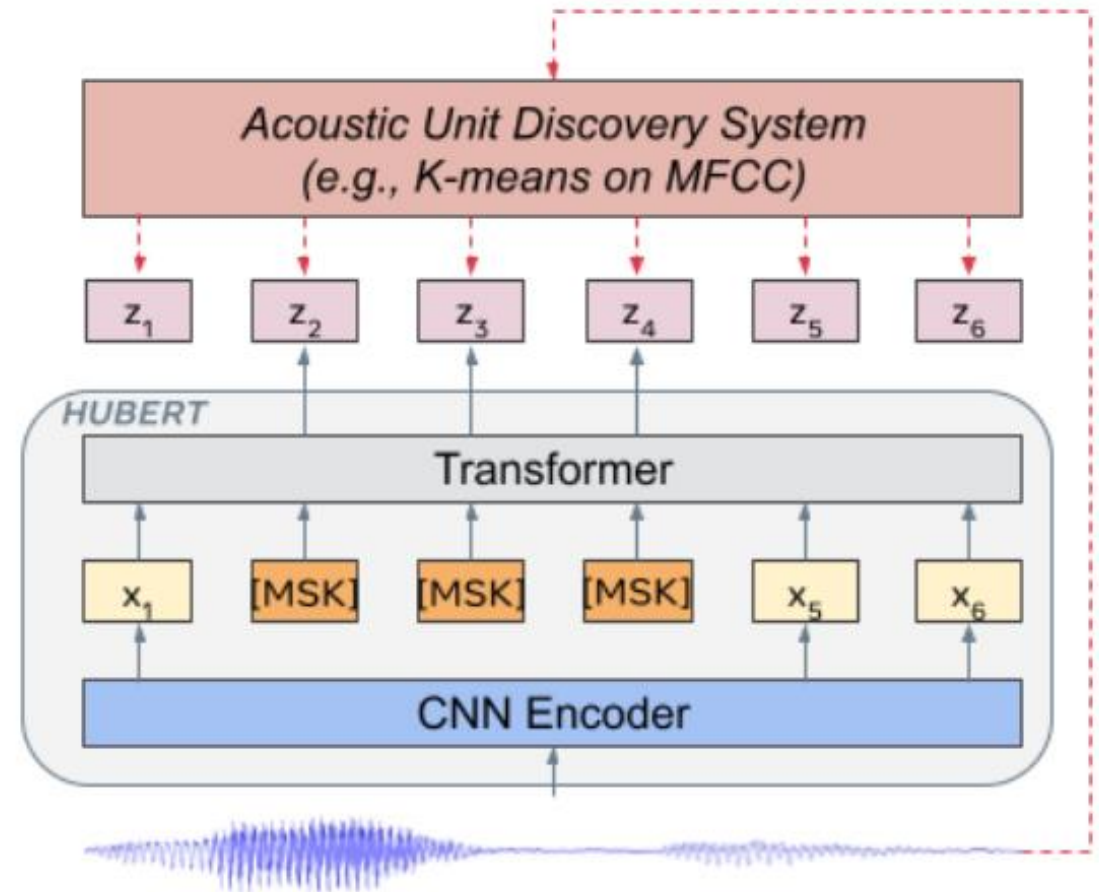
HuBERT: Iterative Quantization

- Create frame-aligned cluster labels offline, then hide spans of the input
- Predict hidden labels from context so the encoder must capture temporal regularities



How HuBERT Builds its Targets

- First iteration: cluster MFCC features using k-means
- Later iterations: cluster hidden states from the previous HuBERT model
- Keep labels fixed within an iteration; train the next predictor against them



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Representation Learning for Understanding Tasks

- Pretraining uses unlabeled LibriSpeech or Libri-Light audio
- For ASR, replace the cluster head and fine-tune with transcribed speech using CTC
- Inference adds a decoding procedure, sometimes with an external language model

Evaluation Metrics

- Key challenge: sequence generation and hierarchical sequence generation

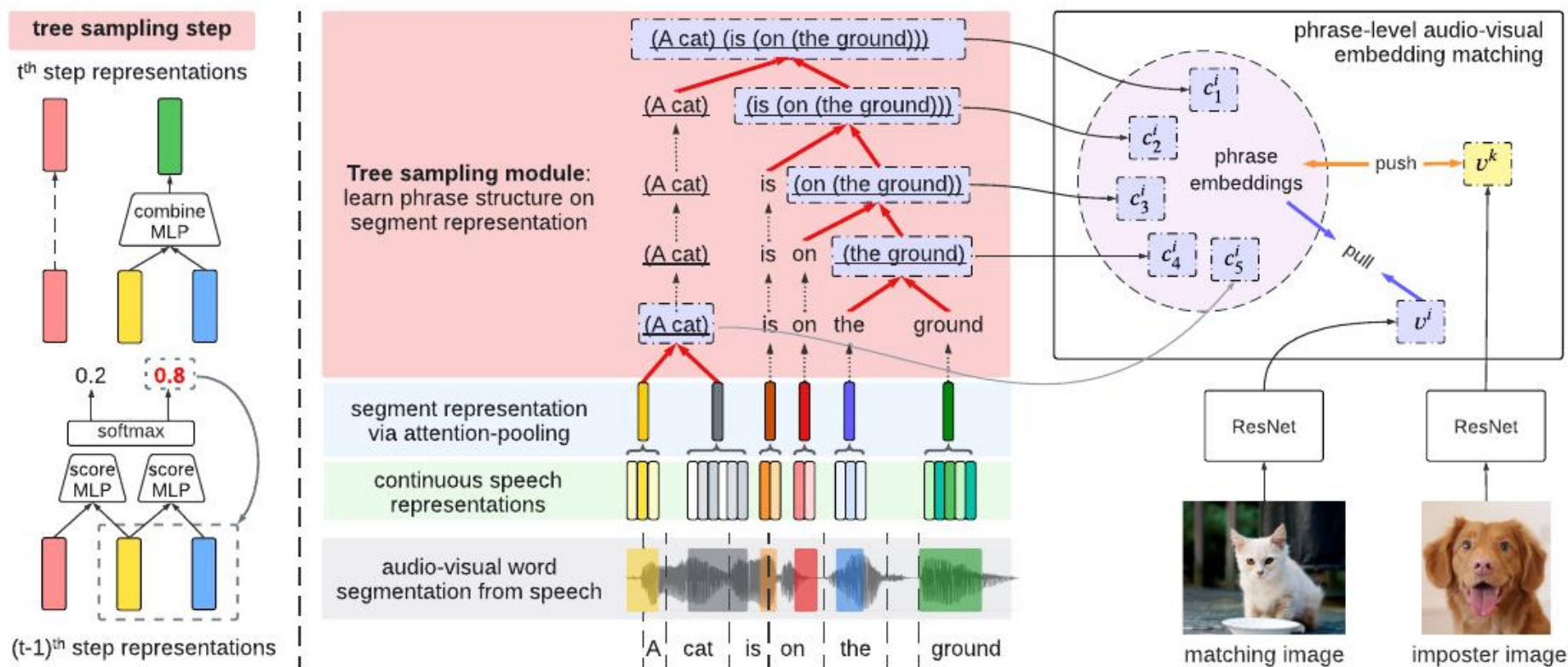
WER for Sequence Generation

- Align a predicted word sequence to the reference
- Count substitutions, deletions, and insertions relative to reference length
- WER ignores timing, voice quality, and whether one error changes the meaning

$$WER = \frac{S + D + I}{N_{ref}} \quad \text{Example: } \frac{0 + 1 + 0}{4} = 25\%$$

Background and mechanism of AV-NSL

- Key idea: merge speech segments into phrases whose embeddings match paired images



Syntax Evaluation with Uncertain Boundaries

- Boundary F1 measures word-segmentation precision and recall
- SAIoU compares temporal constituents under "structural alignment"
- PARSEVAL F1 is suitable when oracle word segmentation fixes the tokenization

What is evaluated?	Metric	What it misses
Word boundaries	Boundary F1	Phrase structure
Trees with audio spans	SAIoU	Full semantic correctness
Trees on fixed words	PARSEVAL F1	Segmentation errors

Word Segmentation Error

- A match allows at most 20 ms of timing error in either direction
- Example: 2 true positives (TP), 2 false positives (FP), 1 miss (FN)

Reference time (s)	Predicted time (s)	Outcome
0.40	0.39	Match: 10 ms error
0.85	0.86	Match: 10 ms error
—	1.05	False positive
1.20	1.23	30 ms: miss + false positive

$$Precision = 50\% \quad Recall = 66.7\% \quad F1 = \frac{2TP}{2TP + FP + FN} = 57.1\%$$

Tree Distance Given Words

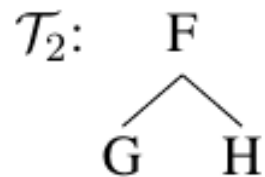
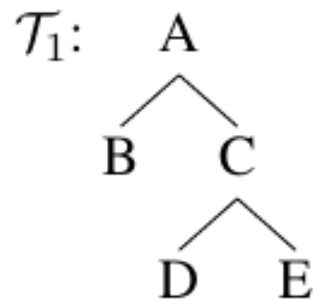
- With fixed words, count exact unlabeled span matches
- Example: “the singer plays the piano” has one shared nontrivial span

Reference constituents	Predicted constituents	Exact match?
the singer [0, 2)	the singer [0, 2)	Yes
the piano [3, 5)	plays the [2, 4)	No
plays the piano [2, 5)	the singer plays the [0, 4)	No

$$P = R = \frac{1}{3} \quad F1 = \frac{2PR}{P + R} = 33.3\%$$

Tree Distance + Word Segmentation Error

- Each node matches at most one node in the other tree
- Matched ancestors and descendants must keep that relation in both trees
- Example: A–G plus E–H is invalid because G is not an ancestor of H



$A \leftrightarrow G$ and $E \leftrightarrow H$: conflicted (violating rule 10.1)

$C \leftrightarrow F$ and $A \leftrightarrow H$: conflicted (violating rules 10.2 and 10.3)

$B \leftrightarrow H$ and $C \leftrightarrow F$: conflicted (violating rule 10.4)

$A \leftrightarrow F$ and $C \leftrightarrow H$: not conflicted

Choosing an evaluation for the task

- Recognition: WER with the label budget and decoder specified
- Restoration/separation: fidelity, perception, leakage, and runtime
- Generation/grounding: control satisfaction, human judgment, and structural validity

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Background: Symbolic Music Models

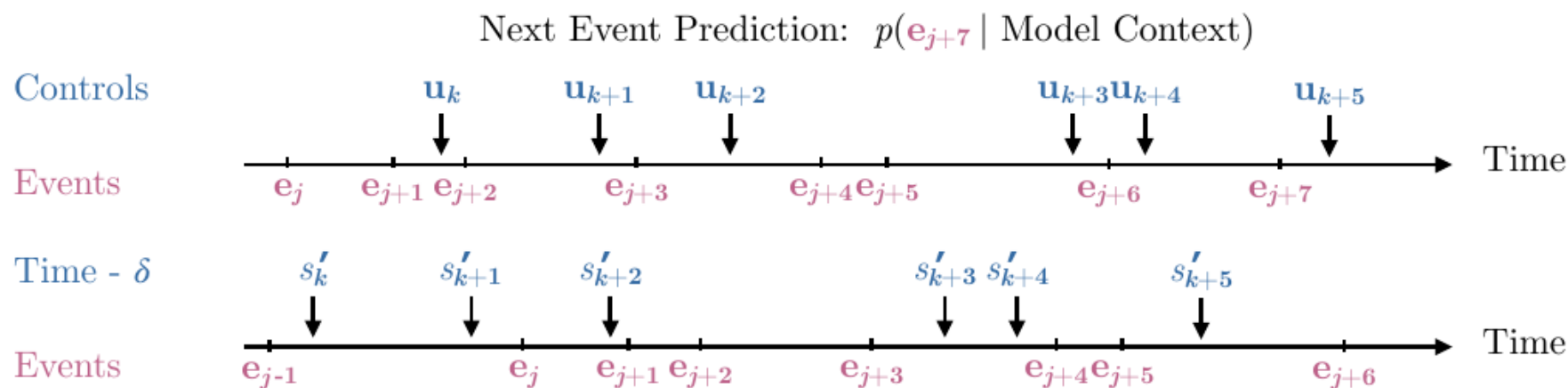


Background: Symbolic Music Models



AMT: the key idea

- Key idea: interleave future controls early enough to condition the next event



Serialized Sequence Structure

Unconditional: $\mathbf{e}_1, \dots, \mathbf{e}_j, \mathbf{e}_{j+1}, \mathbf{e}_{j+2}, \mathbf{e}_{j+3}, \mathbf{e}_{j+4}, \mathbf{e}_{j+5}, \mathbf{e}_{j+6}, \mathbf{e}_{j+7}, \dots$

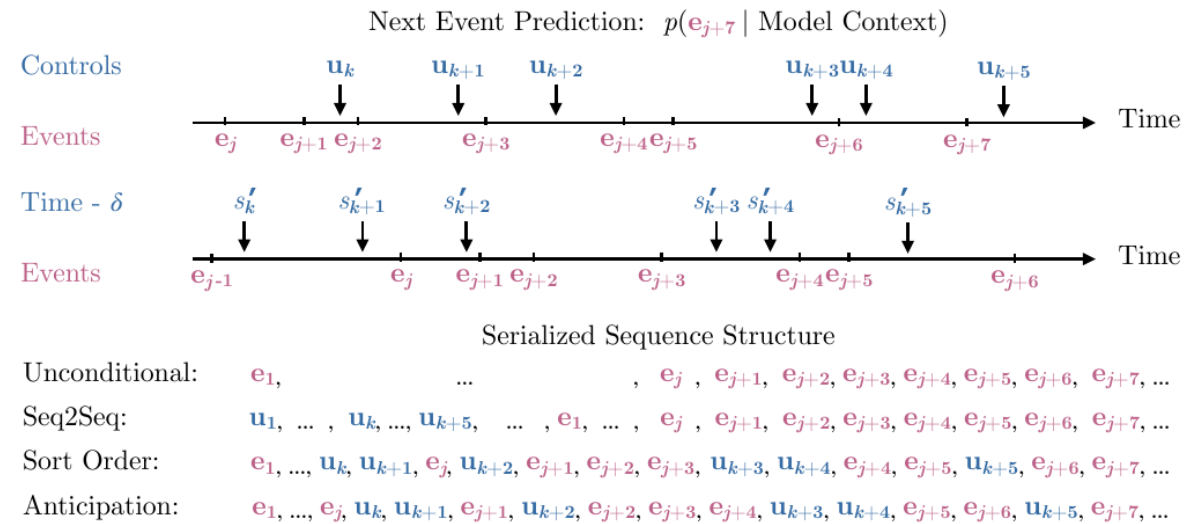
Seq2Seq: $\mathbf{u}_1, \dots, \mathbf{u}_k, \dots, \mathbf{u}_{k+5}, \dots, \mathbf{e}_1, \dots, \mathbf{e}_j, \mathbf{e}_{j+1}, \mathbf{e}_{j+2}, \mathbf{e}_{j+3}, \mathbf{e}_{j+4}, \mathbf{e}_{j+5}, \mathbf{e}_{j+6}, \mathbf{e}_{j+7}, \dots$

Sort Order: $\mathbf{e}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \mathbf{e}_j, \mathbf{u}_{k+2}, \mathbf{e}_{j+1}, \mathbf{e}_{j+2}, \mathbf{e}_{j+3}, \mathbf{u}_{k+3}, \mathbf{u}_{k+4}, \mathbf{e}_{j+4}, \mathbf{e}_{j+5}, \mathbf{u}_{k+5}, \mathbf{e}_{j+6}, \mathbf{e}_{j+7}, \dots$

Anticipation: $\mathbf{e}_1, \dots, \mathbf{e}_j, \mathbf{u}_k, \mathbf{u}_{k+1}, \mathbf{e}_{j+1}, \mathbf{u}_{k+2}, \mathbf{e}_{j+2}, \mathbf{e}_{j+3}, \mathbf{e}_{j+4}, \mathbf{u}_{k+3}, \mathbf{u}_{k+4}, \mathbf{e}_{j+5}, \mathbf{e}_{j+6}, \mathbf{u}_{k+5}, \mathbf{e}_{j+7}, \dots$

Arrival-time tokens preserve meaning

- Each note carries its onset time, duration, and instrument–pitch identity
- Reordering complete note triples preserves their absolute timing information



How AMT learns and preserves controls

- Train on Lakh MIDI with random time spans, instruments, or notes chosen as controls
- Use next-token prediction on the interleaved sequence, including control tokens
- At inference, insert user controls and generate the remaining events

Questions for Discussion

- Could two sources playing the same note and still be separable?
- When prescribed control notes conflict with the model's learned musical prior, should the model obey the user or repair the music?
- Is sound a useful modality in your application of interest? What kind of information do you expect from it?